CHAPTER 3 · CONCEPT 4

Immunity, Vaccines and Treatment

How the body fights germs, how vaccines train it in advance, and why antibiotics must be used wisely.

Defence, prevention and cure

The body's natural ability to fight disease is **immunity**; the **immune system** does this job.

Vaccines — prevention

A **vaccine** trains the immune system to recognise and attack a germ, giving **acquired immunity**. Vaccines may use weakened/dead pathogens or harmless parts of them. They are **preventive, not curative** — they stop disease before it happens but don't treat someone already sick.

> **Edward Jenner (late 1700s):** noticed milkmaids who had had **cowpox** did not get **smallpox**. He used cowpox material to protect a boy — the **first vaccine**. Smallpox was finally eradicated worldwide in 1980. (India's traditional method was **variolation**.)

Antibiotics — cure for bacteria

Antibiotics kill **bacteria** by targeting parts of bacterial cells that differ from our cells. They do **NOT** work against **viruses** (so they don't cure colds/flu) or protozoa.

> **Alexander Fleming (1928):** discovered **penicillin**, the first antibiotic, when a mould stopped bacteria from growing.

Antibiotic resistance

Misusing antibiotics (taking them when not needed, or not finishing the dose) lets bacteria **survive and multiply — antibiotic resistance**. Use antibiotics only when a doctor prescribes them, in the correct dose and full duration.

> **Remember:** vaccines prevent; antibiotics cure bacterial infections; neither cures a viral infection by itself.

Where students lose marks

> **Trap 1 — Antibiotics don't work on viruses.** So they won't cure colds, flu or other viral infections.

> **Trap 2 — Vaccines prevent, they don't cure.** A vaccine protects before infection; it can't treat an existing illness.

Trap 3 — Jenner used cowpox, not smallpox, for the vaccine. Cowpox is the milder, related disease.

Resistance: not finishing the prescribed course or taking antibiotics needlessly causes antibiotic resistance.

Model answers — write them like this

Example A. Why should we not take antibiotics for a cold or flu?

Step 1: Cold and flu are caused by viruses.

Step 2: Antibiotics act only on bacteria, not viruses.

∴ Antibiotics can't kill the virus, and misuse causes resistance.

Example B. Why is the immune response stronger the second time we meet the same germ?

Step 1: The first exposure trains the immune system to recognise the germ.

Step 2: It "remembers" the germ, so it responds faster and stronger next time.

∴ Acquired immunity gives a quicker, stronger second response.

Ready to practise? → Open the worksheet